

Executive Summary: Predictive Modeling for Course Demand and Revenue Forecasting on EduPro

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1. Business Context

EduPro has a lot of data sitting around — course details, instructor profiles, enrollment numbers, revenue figures — but most of it only gets used to look backward at what already happened. This project was about testing whether that same data could also look forward: could it help predict how a course will do before or shortly after launch, rather than just report on it afterward?

2. Business Problem

Right now, there's no structured way to estimate:

1. **Course demand** — how many people will enroll.
2. **Course revenue** — how much money a course will bring in.

Without that, decisions on pricing, course launches, and instructor planning end up leaning on intuition or on eyeballing similar past courses. This project builds two prediction models to give those decisions something more concrete to work from.

3. Project Objectives

- Predict enrollment for a given course.
- Predict revenue for a given course.
- Figure out which course and instructor characteristics actually matter.
- Compare a few different modeling approaches rather than settling on the first one.
- Judge performance properly, not just by eye.
- Package the result into something reusable.
- Put it behind an interface non-technical people can use.
- Turn the numbers into actual recommendations.
- Make sure the dashboard doesn't break on bad input.

4. Data Overview

The source file is **EduPro Online Platform(1).xlsx**:

Dataset	Records
Users	3,000
Teachers	60
Courses	60

After merging course and teacher data, the final modeling table came out to **60 courses across 15 columns**, with nine of those columns used as actual predictive features:

Categorical

CourseCategory, CourseType, CourseLevel, Expertise

Numerical

CoursePrice, CourseDuration, CourseRating, TeacherRating, YearsOfExperience

5. Prediction Targets

EnrollmentCount

Mean 166.67, standard deviation 12.52, ranging from 140 to 196 — a fairly narrow band, which turned out to make this the harder of the two targets to predict well.

CourseRevenue

Mean 15,188.72, standard deviation 25,409.72, ranging from 0 up to 85,416.60. Thirty-eight of the sixty courses had zero recorded revenue, so this target is really a mix of “made nothing” and “made quite a lot,” rather than something evenly spread.

6. Analytical Methodology

The work moved through the usual stages in order: cleaning, merging, exploring the data, engineering features, preparing prediction targets, splitting into train/test sets, testing a baseline, trying progressively better models, picking a final model for each target, building a reusable pipeline, and finally wrapping it all in a Streamlit dashboard with some testing at the end.

7. Model Development

With only 60 rows total, an 80:20 split was used:

Dataset	Observations
Training	48
Testing	12
Total	60

A basic “predict the mean” baseline was tested first, mostly to have a floor to measure real models against:

Enrollment baseline

MAE 10.44, RMSE 11.77, R^2 -0.05

Revenue baseline

MAE 28,413.56, RMSE 36,217.34, R^2 -0.25

From there, Linear Regression and a couple of tree-based models (Random Forest, Gradient Boosting) were tried and compared.

8. Key Modeling Observation

This is probably the most important finding in the whole project: enrollment and revenue behave completely differently as prediction problems.

Linear Regression on enrollment actually did worse than the naive baseline — MAE 14.20, RMSE 19.81, R^2 -1.97. On revenue, the same technique performed extremely well — MAE 2,725.99, RMSE 3,542.47, R^2 0.99.

That gap is worth sitting with rather than glossing over. It suggests revenue has a fairly clean relationship to the available features (worth double-checking there’s no leakage there), while enrollment either needs more informative features or simply doesn’t have a strong linear — or even easily nonlinear — relationship with what’s currently available.

9. Predictive Dashboard

The final solution lives in a Streamlit app (`**D:Analytics Project_Predictive_Modeling.py**`). A user enters course details and gets back:

- Predicted enrollment and revenue
- A plain-language read on expected demand
- A short business recommendation
- A summary of what was entered
- Category-level context
- A look at which features are driving the prediction

10. Business Value

Course launch decisions — predicted demand gives a rough, quantitative starting point when comparing course ideas.

Pricing — predicted revenue is useful context, though it shouldn't override market realities or competitor pricing.

Instructor planning — since teacher rating, experience, and expertise are already in the model, they're a natural lens for thinking about instructor-course fit.

Category strategy — category-level patterns can point toward where it might be worth investing more, or where things need a closer look.

Accessibility — none of this is useful if it stays in a notebook, so the Streamlit interface exists specifically to put it in front of non-technical decision-makers.

11. Feature Importance

The tree-based model gives a ranked view of which features it relies on most. That's genuinely useful for understanding what the model is picking up on, but it's not proof of cause and effect — a feature can be important to the model's predictions without actually being what drives enrollment or revenue in real life.

12. Dashboard Testing and Reliability

Before calling this done, I ran the dashboard through a handful of failure scenarios: valid inputs, missing inputs, bad numeric values, categories the model had never seen, missing model files, and outright prediction failures. The **OneHotEncoder(handle_unknown="ignore")** setting does a lot of quiet work here — it lets the app handle an unfamiliar category gracefully instead of crashing. Everywhere else, the goal was the same: fail with a clear message, not a raw error screen.

13. Key Results

Area	Result
Courses analyzed	60
Modeling features	9
Prediction targets	2
Training observations	48
Testing observations	12
Demand prediction	Implemented
Revenue prediction	Implemented
Category analysis	Implemented
Feature importance	Implemented
Interactive dashboard	Streamlit
Dashboard testing	Completed

Target	Final Model	MAE	RMSE	R ²
EnrollmentCount	Random Forest	12.20	13.27	-0.33
CourseRevenue	Linear Regression	2,725.99	3,542.47	0.99

Worth calling out plainly: the enrollment model, even after trying multiple algorithms, still underperforms a flat average prediction. It's the best of what was tried, not a strong model in absolute terms.

14. Business Recommendations

1. Use predicted demand as one input for prioritizing course launches — not the deciding factor.
2. Bring predicted revenue into pricing conversations alongside market and competitive context.
3. Use instructor characteristics already in the model to inform course-instructor pairing.
4. Compare category-level performance, but treat any single category's numbers as a starting point, not a verdict.
5. Treat feature importance as a clue about what the model is using, not a causal explanation.
6. Push for more data collection generally — nearly every recommendation above gets stronger with a bigger dataset.

15. Limitations

The biggest limitation by far is dataset size — 60 courses total, with only 12 held out for testing. That's a small enough sample that a strong-looking result (like the revenue model's R² of 0.99) deserves a second look rather than immediate trust, and a weak-looking result (enrollment) might improve meaningfully with more data rather than needing an entirely different approach.

16. Future Scope

- A meaningfully larger historical dataset
- Time-based features: launch timing, monthly trends, seasonality, promotions
- Additional algorithms once there's more data to justify them
- Proper hyperparameter tuning
- Ongoing monitoring of predictions versus actual outcomes
- Automated retraining as new data comes in
- Confidence intervals, trend views, and scenario/pricing simulations in the dashboard

17. Executive Conclusion

This project built a working, end-to-end system for predicting course enrollment and revenue on EduPro, and packaged it into a dashboard that turns model output into something a business user can actually act on.

The honest headline is that the two targets aren't equally well served yet — revenue prediction is strong (pending a sanity check on why it's quite this strong), while enrollment prediction still has real room to improve. That's a useful, specific finding on its own, and it gives a clear direction for the next round of work rather than leaving things as a vague “more data would help” conclusion.

Small dataset and all, this gives EduPro a real foundation to build from as more course and transaction history accumulates.

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